THE PERFECTION OF THE STRUCTURE SHEAF IN CHARACTERISTIC P LIES IN THE ESSENTIAL IMAGE OF THE H-TOPOLOGY PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full for the configurations it treats, with the two cohomological inputs recorded at the start and the reduction to explicit short exact sequences of sections of O^{perf} in each case.

Editorial note. The label lemma-h-sheaf-colim-F names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the seven cited inputs are kept as external pointers with their canonical Stacks tags, including the cross-chapter entries of the flatness chapter.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The h-topology comparison morphism epsilon and its pushforward R epsilon_*. Same chapter, the comparison between the h-topology and the fppf topology on (Sch/S), and the criterion for an object to lie in the essential image of R epsilon_*. Supplies the target of the assertion and the criterion that the proof verifies.
- Cohomology of the perfection as a filtered colimit. Cohomology on Sites, the lemma computing H^i of a filtered colimit of sheaves over a collection, applied to O^{perf} = colim_F O. Supplies the identity H^i_{fppf}(X, O^{perf}) = colim_F H^i_{fppf}(X, O) for quasi-compact quasi-separated X.
- Almost blow-up squares and the exactness statements of the flatness chapter. More on Flatness, the one-generator and two-generator examples together with the funny blow-up and h-sheaf-colimit lemmas. Supplies the two configurations whose short exact sequences must be verified for the criterion.

Source reference. Original proposition 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Prerequisite pointer invoked at proof line(s) 23669; it names a

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0EVG: sources/stacks-chow/tags.txt, source lines 18586–18586. Prerequisite pointer invoked at proof line(s) 23673; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0EVH: sources/stacks-chow/tags.txt, source lines 18587–18587. Prerequisite pointer invoked at proof line(s) 23689; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EVJ: sources/stacks-chow/tags.txt, source lines 18589–18589. Prerequisite pointer invoked at proof line(s) 23692; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EVK: sources/stacks-chow/tags.txt, source lines 18590–18590. Prerequisite pointer invoked at proof line(s) 23685; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EVI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 23614–23614. Prerequisite pointer invoked at proof line(s) 23659; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0739: sources/stacks-chow/tags.txt, source lines 9053–9053. Prerequisite pointer invoked at proof line(s) 23667; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let p be a prime number. Let S be a scheme over $\mathbf{F}_p$. Consider the sheaf $\mathcal{O}^{perf} = \operatorname{colim}_F \mathcal{O}$ on $(Sch/S)_{fppf}$. Then $\mathcal{O}^{perf}$ is in the essential image of $R\epsilon_* : D((Sch/S)_h) \to D((Sch/S)_{fppf})$.*

Proof. We prove this using the criterion of Lemma 0EVI. Before check the conditions, we note that for a quasi-compact and quasi-separated object X of $(Sch/S)_{fppf}$ we have

$$H^i_{fppf}(X, \mathcal{O}^{perf}) = \operatorname{colim}_F H^i_{fppf}(X, \mathcal{O})$$

See Cohomology on Sites, Lemma 0739. We will also use that $H^i_{fppf}(X, \mathcal{O}) = H^i(X, \mathcal{O})$, see Descent, Proposition 03DW.

Let A, f, J be as in More on Flatness, Example 0EVG and consider the associated almost blow up square. Since X, X', Z, E are affine, we have no higher cohomology of $\mathcal{O}$. Hence we only have to check that

$$0 \to \mathcal{O}^{perf}(X) \to \mathcal{O}^{perf}(X') \oplus \mathcal{O}^{perf}(Z) \to \mathcal{O}^{perf}(E) \to 0$$

is a short exact sequence. This was shown in (the proof of) More on Flatness, Lemma 0EVK.

Let X, X', Z, E be as in More on Flatness, Example 0EVH. Since X and Z are affine we have $H^p(X, \mathcal{O}_X) = H^p(Z, \mathcal{O}_X) = 0$ for $p > 0$. By More on Flatness, Lemma 0EVJ we have $H^p(X', \mathcal{O}_{X'}) = 0$ for $p > 0$. Since $E = \mathbf{P}^1_Z$ and Z is affine

we also have $H^p(E, \mathcal{O}_E) = 0$ for $p > 0$. As in the previous paragraph we reduce to checking that

$$0 \to \mathcal{O}^{perf}(X) \to \mathcal{O}^{perf}(X') \oplus \mathcal{O}^{perf}(Z) \to \mathcal{O}^{perf}(E) \to 0$$

is a short exact sequence. This was shown in (the proof of) More on Flatness, Lemma 0EVK. □